

Answer as many questions as you can.

- (1) (a) Let W be the subspace of $\mathbb{R}^3$ spanned by the vectors $(1, 1, 0)^t$ and $(0, 1, 1)^t$. Determine the orthogonal projection of the vector $v = (1, 0, 0)^t$ into W , and the orthogonal projection of v into $W^\perp$.
(b) When do we call a square matrix an orthogonal projector? Show that a square matrix Q is an orthogonal projector if and only if $Q^*Q = Q$. (5+5)
- (2) (a) Prove that a square matrix A is unitarily diagonalizable (i.e. there exists a unitary matrix P such that $P^{-1}AP$ is diagonal) if and only if A is normal.
(b) Show that a normal matrix is unitary if and only if every eigenvalue has unit modulus. (10+6)
- (3) Find a singular value decomposition (SVD) of the following matrix:

$$A = \begin{pmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{pmatrix}. \quad (10)$$

- (4) A real $n \times n$ symmetric matrix is called positive definite if $X^TAX > 0$, for all non-zero $X \in \mathbb{R}^n$.
(a) Prove that if A is positive definite then all leading principal submatrices of A have positive determinant.
(b) Conversely, show that if all the leading principal submatrices of a real symmetric matrix A have positive determinant, then A is positive definite. (2+6)
- (5) (a) State Sylvester's law of inertia for real quadratic forms.
(b) Find index and signature of the 3-ary real quadratic form:

$$Q(x_1, x_2, x_3)^t = x_1x_2 + x_2x_3. \quad (2+8)$$

- (6) For any n -ary real quadratic form $Q(X) = X^tAX$, where A is a $n \times n$ symmetric matrix, if S denotes the unit sphere i.e. $S = \{X \in \mathbb{R}^n : \|X\| = 1\}$, then show that

$$\max_{X \in S} Q(X) = \lambda_1, \quad \min_{X \in S} Q(X) = \lambda_n,$$

where λ_1 and λ_n are the largest and smallest eigenvalues of A . (6)
